Royal Devon University Healthcare NHS Foundation Trust

Clinical Guideline for:
Antepartum Haemorrhage
Eastern Services

Summary

This guideline summarises the management of antepartum haemorrhage in all care settings.

Initial Management of Severe APH in all Birth Settings:

- If presents to a Birth Centre transfer to Consultant Led Unit for assessment (Ambulance and midwifery escort is required).
- Ring 2222 and state Obstetric Emergency
- Lie flat with left lateral tilt
- High flow facial oxygen
- Commence IV infusion (2x large bore cannulae)
- Send bloods for FBC, clotting, Kleihauer& cross match 4 units of fresh whole blood.
- If urgent blood transfusion required commence with emergency O- blood from the fridge and call 2222 to instigate the Massive Blood Loss Protocol
- Confirm presence of fetal heart by USS & commence continuous CTG
- Insert urethral catheter and monitor urine output hourly
- Give oral omeprazole
- If immediate delivery by LSCS is not indicated, perform an USS to exclude a placenta praevia
- If praevia not present delivery depends on maternal & fetal condition
    - LSCS, induction as per protocol, or close observation on labour ward.
- Give antenatal steroids if indicated
- If there is any concern for maternal or fetal well-being then delivery should be by LSCS. Consider $MgSO_4$ if <30/40

Contents

1. Definitions 3
1.1 Antepartum Haemorrhage 3
1.2 Severity of Antepartum Haemorrhage 3

2. Prevalence, Risk Factors & Causes 3

3. Management of Minor APH 4

4. Management of Major APH 5

5. Debriefing the Staff & Family 6

6. Training 6

7. References 6

8. Publication Details 6

Appendix 1 - Massive Obstetric Haemorrhage Algorithm 7-8

1.0 DEFINITIONS

1.1 Antepartum Haemorrhage

- Antepartum haemorrhage (APH) is defined as bleeding from or in to the genital tract, occurring from 24^{+0} weeks of pregnancy and prior to the birth of the baby.
- For the purpose of this guideline we will consider any bleeding from 20/40 to be managed as highlighted from section 3.0
- The two main causes of severe APH are placenta praevia and placental abruption.
- It is not uncommon to fail to identify a cause for APH when it is then described as 'unexplained APH' or antepartum haemorrhage of unknown origin (AHUO).

1.2 Severity of Antepartum Haemorrhage

- There are no consistent means by which the severity of APH is classified.
- With a concealed abruption the amount of blood lost may be markedly underestimated.
- The RCOG uses the following definitions:
  - Minor haemorrhage – blood loss less than 50 ml that has settled
  - Major haemorrhage – blood loss of 50–1000 ml, with no signs of clinical shock
  - Massive haemorrhage – blood loss greater than 1000 ml and/or signs of clinical shock

2.0 PREVALENCE, RISK FACTORS & CAUSES

- Prevalence of antepartum haemorrhage of unknown origin (AHUO) is up to 72 %
- Likelihood of antepartum haemorrhage and subsequent delivery before 37 weeks is significant: odds ratio (OR) of 3

2.1 Causes of APH include:

- Placenta Praevia
- Placental Abruption
- Uterine Rupture
- Vasa Praevia

2.2 Placenta Praevia (See Clinical Guideline for placenta praevia)

- Incidence of 0.5% - 1% at term. Increases to 2% in grand multiparity.
- May be identified at first scan, however it is detected in 25% of pregnancies at second trimester scan.
- The earlier the first episode of bleeding the worse the prognosis.
- Risk factors include previous uterine surgery, previous caesarean section, previous placenta praevia and multiple pregnancy.

2.3 Classification

- **Placental Praevia:** The placenta lies over the internal cervical os.
- **Low lying Placenta:** The leading edge of the placenta is ≤ 20mm from the internal os but not covering the cervical os.

2.4 Placental abruption

- Due to the separation of normally situated placenta, where bleeding occurs into the decidua basalis beneath the placenta and forms a haematoma separating the placenta from the maternal vascular system. Bleeding may be concealed or revealed usually causing uterine pain.
- Can occur at any stage of pregnancy.
- Most maternal complications arise from hypovolaemia and collapse of the systems.
- Causes can include: hypertensive disorders such as pre-eclampsia, sudden decompression of the uterus such as rupture of membranes with polyhydramnios, previous history of abruption, trauma, and use of drugs such as cocaine.

2.5 Vasa Praevia

- This is rare occurring in 1 in 1200 to 1 in 5000 pregnancies
- Blood loss from a ruptured vasa praevia will be life threatening to baby but will have no impact on maternal well being
- See Vasa Praevia guideline (hyperlink to vasa praevia guideline).

2.6 **Always be prepared for a post-partum haemorrhage in women who have had an antepartum haemorrhage.** See trust clinical guideline for management of massive haemorrhage.

3.0 MANAGEMENT OF MINOR APH

Assess fetal and maternal condition on the Maternity Triage Unit or Labour ward:

- If > 26/40 measure and plot symphysis-fundal height (if not compatible with gestational range arrange growth scan)
- Check lie / presentation / engagement
- Perform CTG if ≥ 26/40
- Confirm blood group and follow the Management of antenatal women who are Rh D negative or have atypical red cell antibodies guideline as appropriate. If over 20 weeks gestation take Kleihauer
- Review 20 week anomaly scan for placental site to exclude praevia / low lying placenta.
- Perform gentle speculum examination to check for cervical cause of bleeding. If cervix has unusual or suspicious appearance (including polyps) for middle grade/Consultant review and refer to colposcopy if deemed necessary. There is no indication to perform a vaginal swab for infection
- Provided normal examination and no active bleeding / signs of labour consider discharging particularly those women who.
  - Live ≤ 30 minutes of labour ward

  - Minor bleed - no active bleeding
  - No uterine activity
  - USS confirming no placenta praevia.
  - If low risk, treat rest of pregnancy as normal provided there is no further bleeding
- Recommend senior medical review and admission if high risk. If no further bleeding after 24 hours, discharge home with clear documented information as to incident and associated care, and ensure that the women understands all that has occurred during the episode, and symptoms to seek further support and advice.
- Consider antenatal steroids (if <34/40) if significant bleed or risk of pre-term delivery
- Recurrent attendees with 2 episodes or more of bleeding of unknown cause:
  - Arrange growth scan if < 38 completed weeks
  - Either discuss with on call consultant team or refer consultant antenatal clinic for discussion of further management
  - Consider induction at term if not delivered

4.0 Management of Major Antepartum Haemorrhage – Appendix 1

- If the woman presents to a Birth Centre with a severe APH, transfer to the Consultant Led Unit for assessment. (Ambulance and midwifery escort is required) (Maternal Transfer by Ambulance Guideline).
- If a woman telephones with a history of heavy or continuous bleeding advise she calls 999 for ambulance transfer to hospital and inform labour ward coordinator.
- In the event of severe APH ring 2222 and state Obstetric Emergency
- Give high flow oxygen 10-15 l/min – use reservoir bag and mask
- Commence IV infusion (x 2 large bore cannulae).
- Send bloods for FBC, Clotting, Kleihauer and cross match 4 units of fresh whole blood.
- If blood transfusion is required urgently, commence with emergency O –ve blood kept in the fridge and instigate Massive Blood Loss Protocol - ring 2222 and state, 'I am triggering the major obstetric haemorrhage protocol'.
- Where replacement blood products are required and the woman refuses to accept advice and treatment please refer to the Clinical Guideline for women refusing blood or blood products in pregnancy, labour or postnatally. Confirm presence of fetal heart by USS.
- Commence continuous CTG if>26/40.
- Insert urinary catheter record output hourly.
- Give oral omeprazole 20mg as per protocol.
- If immediate delivery by LSCS is not indicated, perform an USS to exclude a placenta praevia or low lying placenta.
- If praevia not present then treatment should be dependent on maternal and fetal condition i.e. LSCS, induction as per protocol or close observation on labour ward
- Give antenatal steroids if indicated
- If there is any concern about maternal or fetal well-being delivery should be by LSCS.

- If <30/40 give MgSo4 infusion- a loading dose can be given even in cases of Cat 1 sections (See Clinical Guideline for management of Pre-Term Labour)

4.1 **Once a major haemorrhage* has been identified, the following should be available:**

- Blood warmer
- Pressure bags for rapid infusion
- TEG & Coagulopathy correction (FFP, platelets, cryoprecipitate, Ca^{2+} as needed)
- Patient warming device i.e. disposable space blankets and/or Bair Hugger
- Cell salvage to be used if surgery
- Following assessment if appropriate (use of O negative blood, requiring >3 unit Xmatched blood or in situations of obvious massive uncontrolled haemorrhage and/or coagulopathy) instigate Massive Obstetric Haemorrhage Protocol

5.0 DEBRIEFING THE STAFF & FAMILY

Major obstetric haemorrhage can be traumatic to the woman, her family and the birth attendants; therefore, debriefing is recommended by a senior member of the team who was involved at the time of events at the earliest opportunity

6.0 TRAINING

All birth attendants must attend annual 'skills drills' training on the management of postpartum haemorrhage.

7.0 REFERENCES

RCOG Green Top Guideline 63 Antepartum Haemorrhage November 2011

8.0 PUBLICATION DETAILS

Author of Clinical Guideline	Labour Ward Lead Consultant
Directorate/Department responsible for Clinical Guideline	Specialist Services, Child and Women's Health, Maternity
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Appendix 1

Massive Obstetric Haemorrhage (Antepartum >24 weeks gestation) –

Definition: Blood loss > 1000 mls (or class III / IV

Call for help

- Dial 2222. Say: "Obstetric Emergency" and state location

ABC assessment

Left lateral position & give oxygen (non-rebreathe mask 15 L/min)

- IV access: 2 large-bore cannulae
- Bloods: FBC, U&Es, LFTs, coag, venous gas; **Crossmatch 4 units RBCs**
- IV Fluid: Warmed Hartmanns
- Auto-transfuse: Head down tilt
- Observations: Pulse, BP, respiratory rate, O2 sats & temperature
- Assess for other signs of shock e.g. agitation / confusion / loss of consciousness, cool pale moist skin
  - **Clinical shock is an important indicator of maternal blood loss. Bleeding may be concealed.**
- Assess abdomen for pain, tenderness & rigidity
- Weigh all swabs & estimate blood lost

Perform CTG if >26 weeks :

Is immediate delivery of baby necessary? (maternal or fetal compromise)

Blood products

- 2222 State "I want to activate the Major Obstetric Haemorrhage Protocol"
- https://royaldevonstaff.nhs.uk/download.cfm?ver=14400
- Initially 4 units red cells (crossmatched, group specific or "O Rh D negative" as appropriate)
- For further details **See blood component box overleaf**

Decide mode of delivery

- Consider cause: Placenta previa, placental abruption, uterine rupture, vasa previa

Other Monitoring & Investigations

- Keep woman warm:

Warmed fluid & bair hugger

- HemoCue and blood gas & TEG
- Urinary catheter & urometer
- Consider central and arterial lines
- Regular Observations

Continue overleaf

Continuation from overleaf

At delivery

- Beware of coagulopathy associated with placental abruption – run TEG
- Are pH, temperature & clotting normal?
- Beware of significant risk of PPH

Postpartum

- Consider HDU / ICU
- Antibiotics if uterus or vagina explored
- **Thromboprophylaxis assessment** when bleeding controlled

Blood component therapy

Red cells, FFP & platelets:

- Initially request **4 units red cells** for major obstetric haemorrhage
- If required, then request **major haemorrhage pack** (**4 units red cells + 2 units FFP**)
- Consider seeking advice from consultant haematologist
- If required then give **major blood loss pack 2**
- For further information see MH flow chart https://royaldevonstaff.nhs.uk/download.cfm?ver=14400
- **Targets:**
- Aim Hb > 80 g/L; platelets >75 x 10^9/L; INR < 1.5; fibrinogen > 200mg/dl

American College of Surgeons Classes of Acute Haemorrhage				
Factors	**1**	**2**	**3**	**4**
Blood Loss (%)	<15	15-30	30-40	>40
Blood Loss (ml)	<750	750-1500	1500-2000	>2000
Pulse	>100	>100	>120	>140
Blood Pressure	Normal	Normal	↓	↓↓
Pulse Pressure	N or ↓	↓	↓↓	↓↓
Capillary refill (s)	<2	2-3	3-4	>5
Respiratory Rate	14-20	20-30	30-40	>40
Urine Output (ml/hr)	≥30	20-30	5-10	Negligible
Mental status	Slightly anxious	Mildly anxious	Anxious / confused	Confused / lethargic

Please remember the impact of different levels of haemorrhage on different maternal booking weights. Smaller patients will be more greatly impacted by a seemingly smaller haemorrhage.